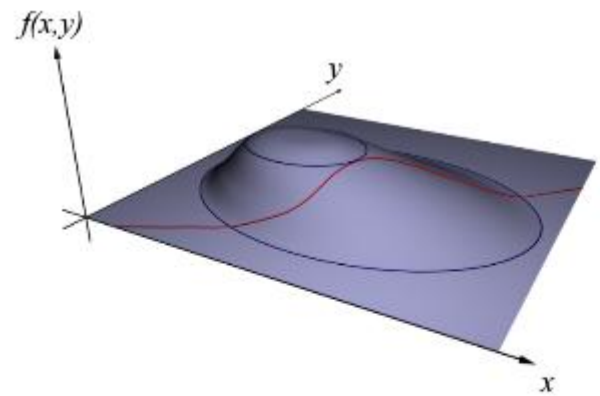
3.9 Lagrange Multipliers

In the last section we looked at finding maximums/minimums of functions of two variables. In this section we are interested in working maximizing/minimizing functions under a given constraint. This section is analogous to the topic of optimization from Calculus I.

Motivation:
Suppose a company makes x units of item A and y units of item B. Suppose also that their profit is given by the function $P(x,y)=7x^2y-3x+18y$. If the company must make exactly 10,000 total units (that is we must have $x+y=10000$) how many units of item A and B should it make to maximize its profit?



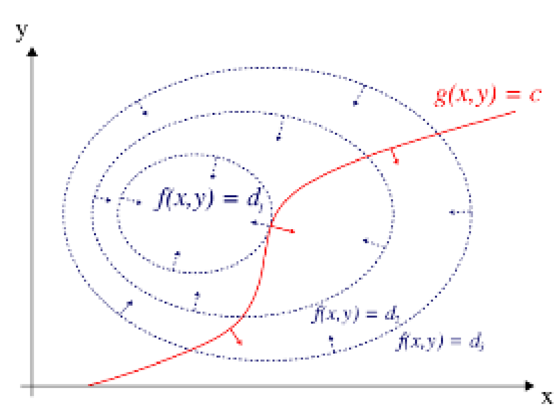
Questions of the above form are the focus of this section. One may correctly notice that the above question can be solved with the techniques of calculus one. However, if our profit function is made up of more than two variables we need a more powerful technique.

We know that $f(x,y)$ forms a surface in 3 space and $x+y=10000$ forms another surface (cutting up from the *xy*-plane along the line $y=-x+10000$). The intersection of the two forms a path along the surface (similar to the image above).

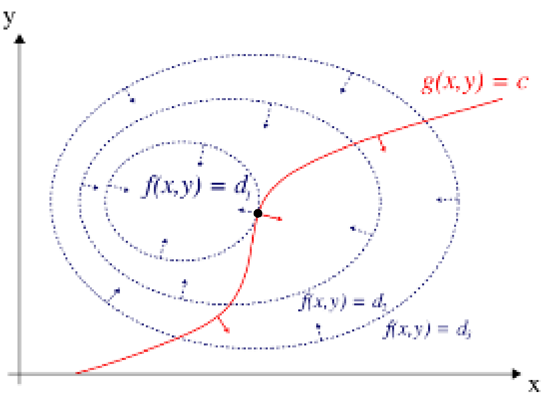
So, if we draw some level curves for $f(x,y)$ (shown above in 3D, or below in the 2D *xy*-plane), it is clearly the case that when the maximum of $f(x,y)$ subject to $g(x,y)=c$ occurs the path $g(x,y)$ brushes against some specific level curve of $f(x,y)$.

Recall that the gradient has a property that it points perpendicular to the surface/curve for which it is associated. Therefore, we can determine where this "brushing up against" occurs by looking at when the gradients of the level curves are parallel to the gradients of the constraint surface.

We show this more formally on the next page:



Let's take a closer look at problems of the form: Find the extreme values of $f(x,y)$ subject to a constraint of the form $g(x,y)=k$ (in hopes of being able to generalize to more variables). We know that $f(x,y)$ forms a surface in 3 space and $g(x,y)=k$ forms a curve in two space. We will assume $f(x,y)$ is differentiable. So, if we draw some level curve for $f(x,y)$ along with the curve for $g(x,y)=k$, we get a picture like:



In the picture it appears that the maximum of $f(x,y)$ subject to $g(x,y)=k$ is d_1 and it occurs at the point highlighted in black.

Suppose we call the point in black (x_0,y_0). Suppose that the curve given by $g(x,y)=k$ has a smooth parameterization: $\mathbf{r}(t)=\langle x(t),y(t)\rangle$. Since (x_0,y_0) is on the curve determined by $g(x,y)=k$, we know that there is a t_0 so that $\mathbf{r}(t_0)=\langle x(t_0),y(t_0)\rangle=\langle x_0,y_0\rangle$.

Now, consider $h(t)=f(x(t),y(t))$ (the idea behind the function h is that we input a point on $g(x,y)=k$ into f). Since f is maximized subject to $g(x,y)=k$ at (x_0,y_0), we know that $h(t_0)$ is a maximum value of h, and $h'(t_0)=0$. (This is because $h(t)$ is on $f(x,y)$ which is differentiable and so $h'(t_0)$ cannot be undefined.)

By the chain rule we know that $\frac{dh}{dt}=\frac{\partial f}{\partial x}\frac{dx}{dt}+\frac{\partial f}{\partial y}\frac{dy}{dt}$.

This means that $0=h'(t_0)=f_x(x_0,y_0)x'(t_0)+f_y(x_0,y_0)y'(t_0)=\vec{\nabla} f(x_0,y_0)\cdot\mathbf{r}'(t_0)$.

So, $\vec{\nabla} f(x_0,y_0)$ is orthogonal to the line tangent to the curve determined by $g(x,y)=k$ at (x_0,y_0).

We also know from our discussion of gradient that $\vec{\nabla} g(x_0,y_0)$ is orthogonal to the line tangent to the curve determined by $g(x,y)=k$ at (x_0,y_0).

So, as long as $\vec{\nabla} g(x_0,y_0)\neq\mathbf{0}$, there is a constant λ such that $\vec{\nabla} f(x_0,y_0)=\lambda\vec{\nabla} g(x_0,y_0)$.

This idea generalizes in higher dimensions and is the main idea behind Lagrange multipliers. We now formally introduce Lagrange multipliers.

Theorem (Method of Lagrange Multipliers):
To find the maximum and minimum values of $f(x,y,z)$ subject to the constraint $g(x,y,z)=k$ (assuming that these extreme values exist and $\vec{\nabla} g \neq \mathbf{0}$ on the surface $g(x,y,z)=k$):

1. Find all values of x, y, z, and λ such that $\vec{\nabla} f(x,y,z)=\lambda\vec{\nabla} g(x,y,z)$ and $g(x,y,z)=k$.
2. Evaluate f at all the points (x,y,z) that result from step 1. The largest of these values is the maximum value of f; the smallest is the minimum value of f.

Note: To find the extreme values of $f(x,y)$ subject to the constraint $g(x,y)=k$, we look for values of x, y, and λ such that $\vec{\nabla} f(x,y)=\lambda\vec{\nabla} g(x,y)$ and $g(x,y)=k$.

Note: If there is only one solution from 1, one may have to test some points to determine whether it is a max or min.

Note: For the method of Lagrange multipliers (as is stated above) we need to assume that the extreme value(s) of interest exists. This will require us to ensure at the conclusion of a problem that our answer(s) make sense. If one wanted to be quite rigorous in using Lagrange multipliers, one would have to give a proof (using other techniques) that the values found in step two are indeed extrema. Since these proof techniques are beyond the scope of our course we will take a more intuitive approach.

<u>Example 1:</u> Find the extreme values of the function $f(x,y)=x^2+2y^2$ on the circle $x^2+y^2=1$.

Using the method of Lagrange multipliers, we must find x, y, and λ such that $\vec{\nabla} f(x,y)=\lambda\vec{\nabla} g(x,y)$ and $x^2+y^2=1$ where $g(x,y)=x^2+y^2$.
We note that $\vec{\nabla} f(x,y)=\lambda\vec{\nabla} g(x,y)\Rightarrow\langle 2x,4y\rangle=\lambda\langle 2x,2y\rangle$. So, we must solve the system:

$$\begin{cases}2x=2\lambda x\\ 4y=2\lambda y\\ x^2+y^2=1\end{cases}\Rightarrow\begin{cases}x=\lambda x\\ 2y=\lambda y\\ x^2+y^2=1\end{cases}\Rightarrow\begin{cases}x(1-\lambda)=0\\ y(2-\lambda)=0\\ x^2+y^2=1\end{cases}.$$

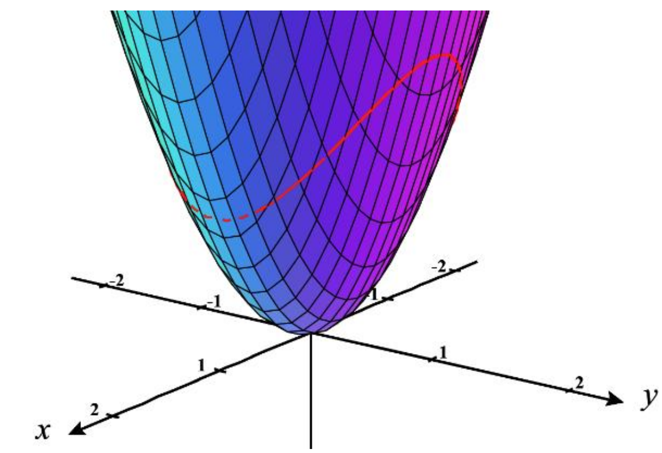
From the first of these equations we obtain: $x=0$ or $\lambda=1$. We explore each case separately.

Consider the case where $x=0$. If $x=0$, then $x^2+y^2=1\Rightarrow y^2=1\Rightarrow y=\pm1$. Then if $y=\pm1$, $y(2-\lambda)=0\Rightarrow\lambda=2$. So, if we write solutions in the form (x,y,λ), solutions to the system include $(0,1,2)$ and $(0,-1,2)$.

Now, consider the case where $\lambda=1$. If $\lambda=1$, then $y(2-\lambda)=0\Rightarrow y=0$. Then if $y=0$, $x^2+y^2=1\Rightarrow x^2=1\Rightarrow x=\pm1$. So, if we write solutions in the form (x,y,λ), solutions to the system include $(1,0,1)$ and $(-1,0,1)$.

Now that we have solved the system, we see that four different solutions are candidates for extreme values:
$f(0,1)=2,\ f(0,-1)=2,\ f(1,0)=1,$ and $f(-1,0)=1$.

So, the maximum value of f on the unit circle is 2, and it occurs at $(0,1)$ and $(0,-1)$.
The minimum value of f on the unit circle is 1, and it occurs at $(1,0)$ and $(-1,0)$.

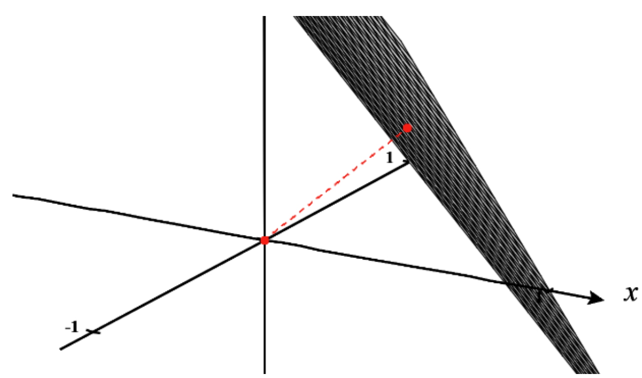


Note: The method of Lagrange Multipliers requires you to solve systems of nonlinear equations. Sadly, there is no algorithmic way to do this (unlike with systems of linear equations, which can always be reduced through elimination/substitution). Some nonlinear equations don't even have a closed expression for their solutions sets, although we will not be dealing with problems like that in this class.

Nonetheless, one may have to be creative when looking for a solution or setting up a problem to avoid creating a tangled mess of equations. The next example illustrates how an adjustment to our setup, or a creative step in our solution, can be highly impactful on our productivity through the problem.

<u>Example 2:</u> Use Lagrange Multipliers to find the shortest distance from the origin to the plane $x+y+z=1$.

We wish to minimize the function
$d(x,y,z)=\sqrt{(x-0)^2+(y-0)^2+(z-0)^2}$ such that $x+y+z=1$.
We know that we can apply Lagrange multipliers to such a problem by finding all values of x, y, z, and λ such that $\vec{\nabla} d(x,y,z)=\lambda\vec{\nabla} g(x,y,z)$ and $x+y+z=1$ where $g(x,y,z)=x+y+z$.



However, since finding $\vec{\nabla} d(x,y,z)$ appears to be a bit tedious, a much more elegant way to do this problem would be to minimize $D(x,y,z)=x^2+y^2+z^2$ such that $x+y+z=1$. This is because we only need to minimize the square of the distance to know the minimum distance.

So, we will plan to proceed by trying to minimize the equation $D(x,y,z)=x^2+y^2+z^2$ subject to the constraint of $x+y+z=1$.

We note that $\vec{\nabla} f(x,y)=\lambda\vec{\nabla} g(x,y)\Rightarrow\langle 2x,2y,2z\rangle=\lambda\langle 1,1,1\rangle$. So, we must solve the system:

$$\begin{cases} 2x = \lambda \cdot 1 \\ 2y = \lambda \cdot 1 \\ 2z = \lambda \cdot 1 \\ x + y + z = 1 \end{cases} \Rightarrow \begin{cases} x = \dfrac{\lambda}{2} \\ y = \dfrac{\lambda}{2} \\ z = \dfrac{\lambda}{2} \\ x + y + z = 1 \end{cases}$$

If we use the first three equations to substitute into the 4th equation and we get that

$$\left(\frac{\lambda}{2}\right)+\left(\frac{\lambda}{2}\right)+\left(\frac{\lambda}{2}\right)=1 \;\rightarrow\; \frac{3\lambda}{2}=1 \;\rightarrow\; \lambda=\frac{2}{3}$$

Therefore, we have that $x=\frac{1}{3},\ y=\frac{1}{3},\ z=\frac{1}{3}$ and this means that the point $\left(\frac{1}{3},\frac{1}{3},\frac{1}{3}\right)$ is the only extrema for the distance from $(0,0,0)$ to the plane $x+y+z=1$.

We calculate the distance from $\left(\frac{1}{3},\frac{1}{3},\frac{1}{3}\right)$ to $(0,0,0)$ to be

$$d=\sqrt{\left(0-\frac{1}{3}\right)^2+\left(0-\frac{1}{3}\right)^2+\left(0-\frac{1}{3}\right)^2}=\sqrt{\left(\frac{1}{9}\right)+\left(\frac{1}{9}\right)+\left(\frac{1}{9}\right)}=\sqrt{\frac{1}{3}}$$

However, we don't know if this would correspond to a maximum of minimum distance.
To check this, we select another point that exists on the plane (say $(1,0,0)$) and compute the distance between this point and $(0,0,0)$.
We see that this distance is $d=\sqrt{(1-0)^2+(0-0)^2+(0-0)^2}=\sqrt{1}=1$.

This clearly shows us that the distance $\sqrt{\frac{1}{3}}$ (between $\left(\frac{1}{3},\frac{1}{3},\frac{1}{3}\right)$ and $(0,0,0)$) is a minimum distance.

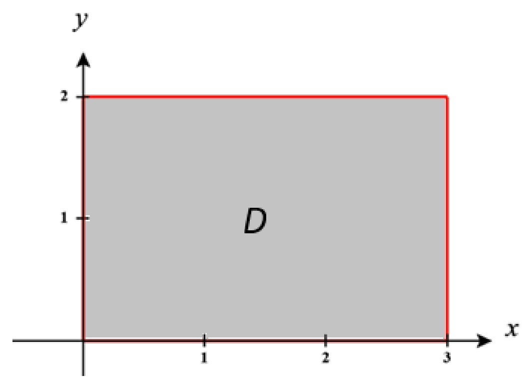
Question: Why is there no point with maximum distance?

Note: This question also could have been solved using the ideas of projections and components as discussed back in Chapter 1.

Example 3: In section 3.8 we tried to find the minimum and maximum value of $f(x,y) = x^2 - 2xy + 2y$ over the rectangle region $D = \{(x,y) | 0 \le x \le 3 \ \& \ 0 \le y \le 2\}$.

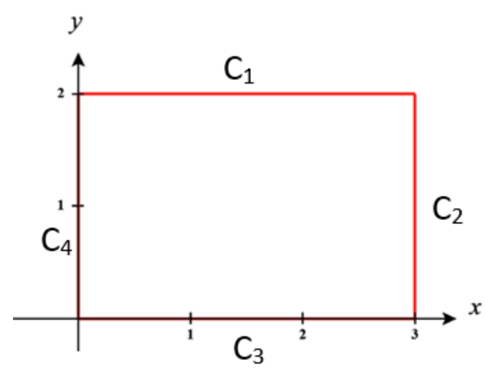
a) Why doesn't the method of Lagrange Multipliers help fully answer this question?

In section 3.8 we were looking for minimums and maximums over the boundary of the region AND the interior of the region. Langrage multipliers would only be able to help look for minimums and maximums along the boundary.



b) Why is the method of Lagrange Multipliers not optimal for checking for minimums and maximums along the boundary of *D*?

We see here that four curves are necessary to fully describe the boundary of *D*. As such, we would have to do four separate Lagrange Multiplier problems (one for each edge).

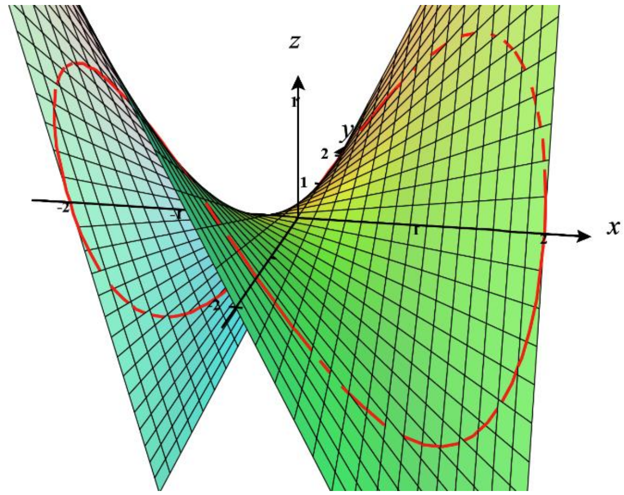


Note: It is important to understand the difference between finding minimums and maximums over closed and bounded sets and the method of Lagrange Multipliers. Although a question may not state directly to use Lagrange Multipliers we have to be able to understand if this technique or the techniques of section 3.8 are required.

Note: Example 1 from this section could have also been completed using the techniques from section 3.8 of finding absolute minimums and maximums on a closed and bounded set, although we would only be checking the boundary of the region determined by the circle and not the full region.

Example 4: Find the maximum and minimum values of $f(x,y)=xy$ subject to the constraint $x^2+2y^2=4$.

We wish to maximize the function $f(x,y)=xy$ along the curve $x^2+2y^2=4$.
So, we want to find all values of x, y, and λ such that $\vec{\nabla} f(x,y)=\lambda\vec{\nabla} g(x,y)$ and $x^2+2y^2=4$ where $g(x,y)=x^2+2y^2$.



We note that $\vec{\nabla} f(x,y)=\lambda\vec{\nabla} g(x,y)\Rightarrow\langle y,x\rangle=\lambda\langle 2x,4y\rangle$.
So, we must solve the system:

$$\begin{cases} y=2x\lambda \\ x=4y\lambda \\ x^2+2y^2=4 \end{cases} \Rightarrow \begin{cases} \dfrac{y}{2x}=\lambda \\ \dfrac{x}{4y}=\lambda \\ x^2+2y^2=4 \end{cases}$$

If we use the first two equations, we can say that

$$\frac{y}{2x}=\frac{x}{4y} \;\to\; 4y^2=2x^2 \;\to\; y^2=\frac{1}{2}x^2.$$

Therefore, by the 3rd equation, we have

$$x^2+2\left(\frac{1}{2}x^2\right)=4 \;\to\; 2x^2=4 \;\to\; x=\pm\sqrt{2}.$$

If $x=\sqrt{2}$ then $y^2=1 \to y=\pm1$. We have possible extrema at the points $\left(\sqrt{2},1\right)$ & $\left(\sqrt{2},-1\right)$.

If $x=-\sqrt{2}$ then $y^2=1 \to y=\pm1$. We have possible extrema at the points $\left(-\sqrt{2},1\right)$ & $\left(-\sqrt{2},-1\right)$.

We calculate that

$$f\left(\sqrt{2},1\right)=\sqrt{2}$$
$$f\left(\sqrt{2},-1\right)=-\sqrt{2}$$
$$f\left(-\sqrt{2},1\right)=-\sqrt{2}$$
$$f\left(-\sqrt{2},-1\right)=\sqrt{2}$$

Therefore, $f(x,y)$ has a maximum of $\sqrt{2}$ and a minimum of $-\sqrt{2}$.

Note: The method of section 3.8 here could be used here to check along the boundary of the region determined by the constraint curve, however this will be more difficult to set up than using Lagrange Multipliers here due to the nature of the functions/equations in this example.

Sometimes the process of solving systems of non-linear equations requires that we take our time, try to organize our work, and systematically work through the possibilities for each implication of our system. The next example demonstrates this.

Example 5: Suppose that a rectangular box without a lid is to be made from 12m^2 of cardboard. Find the maximum volume of such a box.

Suppose that *x*, *y*, and *z* are the length, width, and height, respectively, of the box in meters.

We know that the volume of the box is given by $V(x,y,z)=xyz$ and the surface area of the box is given by $S(x,y,z)=2xz+2yz+xy$.

So, we wish to maximize $V(x,y,z)=xyz$ subject to $2xz+2yz+xy=12$. According to the method of Lagrange Multipliers, we must find *x*, *y*, *z*, and λ such that:

$$\vec{\nabla}V(x,y,z)=\lambda\vec{\nabla}S(x,y,z) \text{ and } 2xz+2yz+xy=12$$

$$\Rightarrow \langle yz,xz,xy\rangle=\lambda\langle 2z+y,2z+x,2x+2y\rangle \text{ and } 2xz+2yz+xy=12.$$

This means that we must solve the nonlinear system:

$$\begin{cases} yz=\lambda(2z+y) \\ xz=\lambda(2z+x) \\ xy=\lambda(2x+2y) \\ 2xz+2yz+xy=12 \end{cases}$$

If we work with the three equations in the above system, we can obtain:

$$\begin{cases} xyz=\lambda x(2z+y) \\ xyz=\lambda y(2z+x) \\ xyz=\lambda z(2x+2y) \end{cases}$$

From the first two of these new equations we obtain:

$\lambda x(2z+y)=\lambda y(2z+x) \quad \Rightarrow \quad \lambda\left[x(2z+y)-y(2z+x)\right]=0$.

This could be true if $\lambda=0$ or if $x(2z+y)-y(2z+x)=0$

We note from the original system that if $\lambda=0$, then $yz=xz=xy=0$ and the equation: $2xz+2yz+xy=12$ would be violated.

So if $x(2z+y)-y(2z+x)=0 \;\Rightarrow\; 2xz-2yz=0 \;\Rightarrow\; 2(xz-yz)=0 \;\Rightarrow\; xz=yz$. This implies that either $z=0$ *or* $x=y$.

We note that if $z=0$, then the first equation of the original system becomes $0=\lambda y$, and since $\lambda\neq 0$, this equation implies that $y=0$.

However, if $z=0$ and $y=0$, $2xz+2yz+xy=12$ would be violated.

So, we know that $z\neq 0$ and $xz=yz\Rightarrow x=y$.

So, we know that

$$\begin{cases} xyz = \lambda y(2z+x) \\ xyz = \lambda z(2x+2y) \end{cases} \Rightarrow \begin{cases} x^2z = \lambda(2zx+x^2) \\ x^2z = \lambda(4zx) \end{cases}$$

The later equations imply $\lambda(2zx+x^2) = \lambda(4zx) \Rightarrow 2zx+x^2 = 4zx \Rightarrow 2zx = x^2$. So $x=0$ *or* $x \neq 0$

We note that if $x=0$, then the second equation of the original system becomes $0 = 2\lambda z$, and since $\lambda \neq 0$, this equation implies that $z=0$. However, if $z=0$ and $x=0$, $2xz+2yz+xy=12$ would be violated. So, we know that $x \neq 0$ and $2zx = x^2 \Rightarrow z = \frac{x}{2}$.

Since $y = x$ and $z = \frac{x}{2}$, we know that $2xz+2yz+xy=12 \Rightarrow x^2+x^2+x^2=12 \Rightarrow x^2=4 \Rightarrow x=\pm 2$. In the context of the problem *x*, *y*, and *z* are nonnegative. So, the only values of *x*, *y*, and *z* that produce a solution to the system are $x=2$, $y=2$, and $z=1$ (the value of λ is one half for these values of *x*, *y*, and *z*).

So, we conclude that maximum volume of a rectangular box without a lid made from 12 m^2 of cardboard is $(2)(2)(1) = 4 \text{ m}^3$ and the maximum volume is achieved with a length and width of 2 m and a height of 1 m (we know that the result is a maximum since x = 3, y = 4, and z = 0 would satisfy $2xz+2yz+xy=12$, but would make $V=0$).